"Practice Midterm Exam"

Problem 1.

a) Express the following complex number in the form $x + yi$ for $x, y \in \mathbf{R}$.

$$\frac{1+3i}{3+i}.$$

b) Express the following complex number in the form $re^{i\theta}$ for $r > 0, \theta \in [0, 2\pi)$.

$$\frac{1+i}{\sqrt{2}}.$$

c) For a complex number $z = x + yi$ for $x, y \in \mathbf{R}$ define the complex conjugate $\bar{z}$ and the modulus $|z|$.

d) Prove that

$$z\bar{z} = |z|^2.$$

Problem 2. Suppose V is a vector space over $\mathbf{F}$.

a) Write the definition that V is a finite dimensional vector space over $\mathbf{F}$.

b) Write the definition that $\{v_1, \dots v_k\}$ is linearly independent.

c) Suppose $\{v_1, \dots v_k\}$ is linearly independent but $\{v_1, \dots v_{k+1}\}$ is linearly dependent. Show that

$$\text{span}\{v_1, \dots v_k, v_{k+1}\} = \text{span}\{v_1, \dots v_k\}.$$

d) Suppose $\{v_1, \dots v_k\}$ is linearly independent in a finite dimensional vector space V. Show that we can extend it to a basis of V.

Problem 3. Suppose we have vector spaces V and W over $\mathbf{F}$ and a linear map $T : V \to W$.

a) Define null(T) and range(T).

b) Define that T is invertible.

c) Show that if T is invertible then $\text{null}(T) = \{0\}$ and $\text{range}(T) = W$.

d) Show that if $\mathrm{null}(T) = \{0\}$ and $\mathrm{range}(T) = W$ then T is invertible. Hint: fix a basis $\{v_i\}$ of V and consider what are $\{T(v_i)\}$ in W.

e) Suppose $V = \mathbf{R}^4$ and $W = \mathbf{R}^3$

$$T : \begin{pmatrix} x \\ y \\ z \\ w \end{pmatrix} \mapsto \begin{pmatrix} x + z - w \\ x + y + w \\ y - z + 2w \end{pmatrix}.$$

Then, find $\dim(\mathrm{null}(T))$.